

ALEXANDER SEBASTIAN CHRISTIANTO

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EDUCATION

Nanyang Technological University, Singapore

Jul 2022 – Jun 2026

Bachelor of Engineering in Electrical & Electronic Engineering, Honours (Distinction) — Specialization in Data Analysis & Machine Learning, Minor in Business

CGPA: 4.04 / 5.0

Relevant Coursework: Artificial Intelligence & Data Mining, Machine Learning Design & Application, Semiconductor Fundamentals, Analog Electronics, Business Finance, Fundamentals of Management, Marketing

CERTIFICATIONS & TECHNICAL SKILLS

Certifications: Google Data Analytics Specialization (Coursera); Google Business Intelligence Specialization (Coursera); Business Intelligence and Data Analytics: Generate Insights — Macquarie University (Coursera)

Technical Skills: Python, SQL, R, SAS Viya, Tableau, Power BI, Looker Studio, Microsoft Excel

Languages: English (Fluent), Indonesian (Fluent), Malay (Fluent), Chinese (Basic)

PROFESSIONAL EXPERIENCE

Data Analyst Intern, Systems on Silicon Manufacturing Company (SSMC)

Jan 2025 – May 2025

- Evaluated and verified incoming material quality for the Supplier Quality Management team, ensuring strict adherence to semiconductor manufacturing standards.
- Analyzed supplier performance metrics and material quality compliance data using Excel, SQL, and Tableau to surface trends for cross-functional review.
- Designed and deployed interactive Tableau dashboards for real-time performance monitoring, improving data visibility for multiple cross-functional teams company-wide.

LEADERSHIP EXPERIENCE

Honorary General Secretary, NTU Christian Fellowship

May 2024 – May 2025

- Partnered with club management to oversee day-to-day operations and shape strategic organizational decisions.
- Streamlined club administration by managing internal communications, coordinating schedules, and maintaining comprehensive meeting documentation to ensure seamless operations.

Freshmen's Orientation Group Leader, NTU

Sep 2022 – Aug 2023

- Led a group of 16 freshmen and 11 committee members through a four-day orientation camp.
- Planned and executed bonding events that fostered enduring friendships among participants.

PROJECTS

Cyclistic Bike-Share Analysis — Business Intelligence & Data Analytics Project

Mar 2026 – May 2026

- Engineered an end-to-end business intelligence solution analyzing millions of NYC Citi Bike trip records to identify customer demand patterns and inform strategic station expansion.
- Authored Stakeholder, Project Requirements, and Planning documents to define success metrics and align cross-functional leadership on business objectives.
- Developed SQL queries and executed ETL processes over a 6-week project lifecycle, integrating primary trip data with secondary Census boundary and weather datasets.
- Designed an accessible, interactive Tableau dashboard to visualize key performance indicators for stakeholders.

AI-Driven Topological Structured Light for Advanced Optical Communication — Final Year Project

Aug 2025 – May 2026

- Developed AI algorithms to optimize the generation of diverse topological light structures using spatial light modulators (SLMs).
- Built neural network models for real-time prediction and compensation of atmospheric turbulence effects on OAM beams.
- Developed a real-time simulator for an adaptive sender AI that automatically switches from multiplexed to single-mode transmission when link health drops below a 95% threshold.

Other Data Analytics & BI Projects

- Hotel Ratings & Trip Type Analysis — Analyzed customer feedback through data visualizations.
- Instagram Clone Database — Performed advanced analysis on a relational database model using SQL.
- NBA 2020 Season Analysis — Cleaned and transformed data to ensure a smooth analysis process.
- Premier League Match Analysis — Investigated match-level data to uncover insights on team performance.

Full write-ups available in portfolio: github.com/alexandersc22/portfolio